

1. (a) A progressive wave is given by equation:

$$y(x, t) = 15 \sin 2\pi(0.5t + x)$$

Where x and y are in cm and t in seconds. Calculate:

- (i) Amplitude
- (ii) Frequency
- (iii) Wavelength
- (iv) Speed of the wave
- (v) Direction of the wave propagation

[10 marks]

- (b) A stationary wave is given by equation:

$$y = 2.5 \cos \pi x \sin 5\pi t$$

Where x and y are in meters and t in seconds. Calculate distance between two nodes.

[4 marks]

- (c) A stretched string of length 100 cm is under tension of 600 N. Calculate the wave speed if its weight is 3.0 g.

[2 marks]

2. (a) A tungsten wire which has diameter, 2.66 mm transfer a charge of 300 C in 65 minutes. If this two meter long wire are connected to 15 V, determine:

- (i) Current in the wire
- (ii) Resistance of the wire
- (iii) Resistivity of the wire

[5 marks]

- (b)

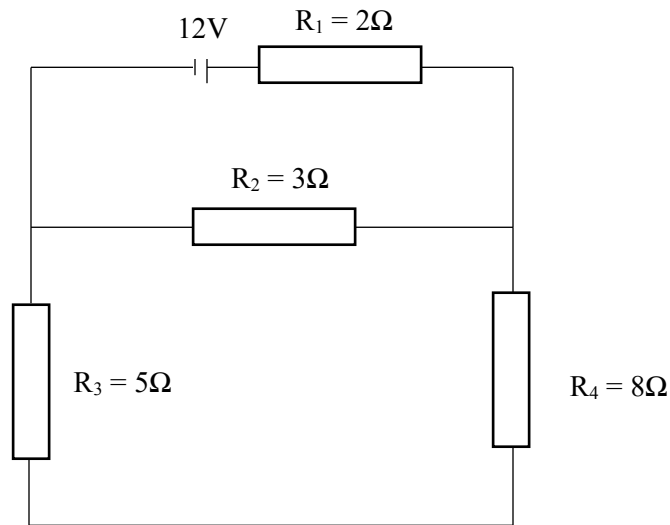


FIGURE 1

FIGURE 1 shows resistors R_1 , R_2 , R_3 , and R_4 are connected to a battery 12V. Calculate

- (i) Effective resistance of the circuit.
- (ii) Total current in the circuit
- (iii) Voltage across resistor R_1

[7 marks]

3. (a)

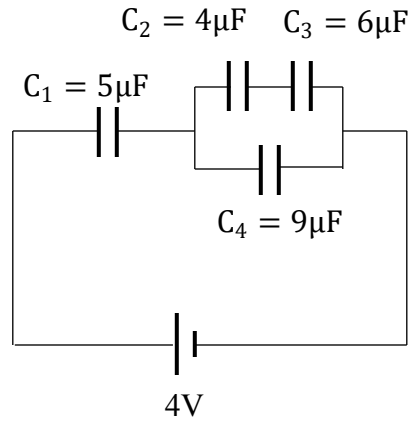


FIGURE 3.1

FIGURE 3.1 shows four capacitors connected in series and parallel. Calculate the

- (i) effective capacitance.
- (ii) charge stored in capacitor C_1 .

[4 marks]

(b)

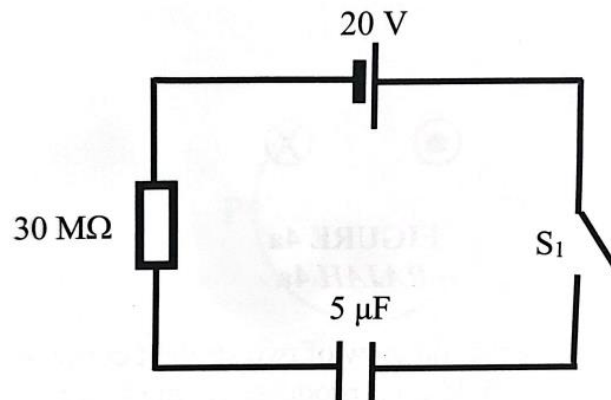


FIGURE 3.2

FIGURE 3.2 shows a direct current circuit. When switch S_1 is closed, the capacitor is charging.

- (i) Calculate the time constant
- (ii) Calculate the time taken for the charge to reach 25% of its maximum value.
- (iii) Sketch a graph of current against time.

[6 marks]

4. a) A solenoid of 25 turns is carrying a current of 50 mA. If the length of the solenoid is 0.8 cm. Calculate the magnetic flux density ~~at the end~~ of the solenoid.

[2 marks]

- b) **FIGURE 4.1** below shown two wires I_1 and I_2 carry currents of 5.0 A in opposite directions and are separated by 8 cm. Calculate the net magnetic field strength at points A, if points A at midway between two wires.

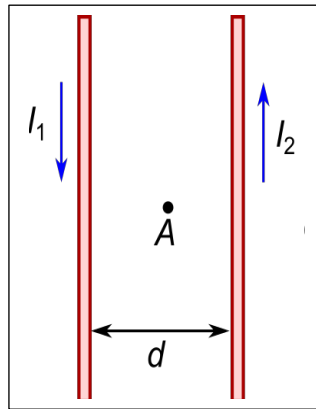


FIGURE 4.1

[3 marks]

- c) A 40 turns coil with diameter of 5.3 cm is placed in a uniform magnetic field of 2.5 T. Calculate the magnetic flux linkage in the coil if the coil makes an angle of 30° to the magnetic field.

[4 marks]

- d) **FIGURE 4.2** shows a square coil of 100 turns with a cross sectional area 3mm^2 is placed perpendicularly to a magnetic field of 5 mT. After 3.0 s, the coil was pulled out between point A and C until the area is reduced to zero.

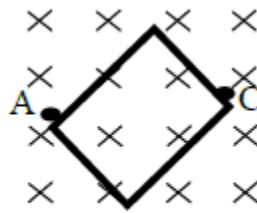


FIGURE 4.2

- i) Calculate the induced emf
- ii) State the direction of induced current

[3 marks]

- e) **FIGURE 4.3** below shows a metal rod is free to slide on horizontal conducting rails that are connected to 5Ω resistor. The system lies in a uniform magnetic field 1.6 T pointing into the page. The rod is moved with a constant velocity $v=3\text{ms}^{-1}$ to the right. Calculate the induce emf.

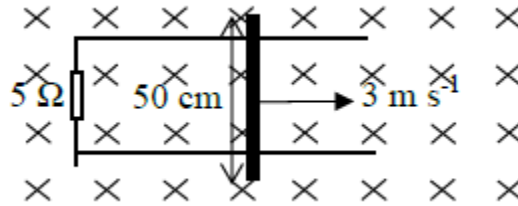


FIGURE 4.3

[2 marks]

5. a) A real object is placed 24 cm from an upright image by using the diverging mirror. The image's height is one fourth of object's height.

- i. Determine the location of the image distance.
- ii. Calculate the radius of curvature of the mirror and describe the type of mirror
- iii. Sketch and label a ray diagram to show the formation of the image

[9 marks]

- b) An object is placed at a distance of 10 cm from a converging lens and real image of linear magnification 3 is formed. Calculate the

- i. Image distance
- ii. Focal of the lens

[5 marks]

6. (a) In a double slit experiment, a light of wavelength 700 nm is incident on the double slit 2.4 m from the screen. The seventh dark fringes pattern is observed 2.5 cm from the central bright.

- (i) Calculate the slit separation of the double slit.
- (ii) Calculate the distance between fifth bright fringes from central maxima.
- (iii) If the experiment is replaced with an unknown wavelength, λ , its second bright fringe is found to be overlapped with the first order bright fringe of wavelength 700 nm. Calculate the unknown wavelength.

[8 marks]

- (b) A blue light of wavelength 460 nm is incident on double slit with separation distance 0.1 mm. The screen is placed 1.5 m from the slit.

- (i) Calculate the separation distance between two consecutive bright fringes.
- (ii) If the blue light is replaced with a yellow light ($\lambda = 580$ nm), state the effect of the fringe separation. Explain your answer.
- (iv) If the separation distance of the double slit increase, state the effect to the interference pattern. Explain your answer.

[6 marks]